
DEEPSEEKER: A FIELD-EVALUATED EMBEDDED CATAMARAN FOR SYNCHRONIZED SHALLOW-WATER IMAGING AND COVERAGE GUIDANCE AT LAKE ISSYK-KUL

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Young Scientists Camp 2026, Lake Issyk-Kul, Kyrgyz Republic

Original Research Article · Field dataset M0016 · 18 July 2026

Abstract

Shallow littoral zones are difficult to inspect repeatedly with conventional survey vessels, yet they are the part of a lake most exposed to recreation, shoreline activity, and visible submerged debris. This paper presents DeepSeeker, a locally developed differential-thrust catamaran designed to collect spatially organized underwater imagery in the nearshore zone of Lake Issyk-Kul. The platform separates navigation and imaging between a Main ESP32 and an ESP32-S3-CAM equipped with an OV3660 sensor. GNSS, inertial, motor, and capture metadata are transmitted over UART; each QXGA JPEG is written to microSD together with a telemetry row and a geographic record. The complete raw archive from mission M0016 contains 208 images, 208 telemetry records, and 208 geotag entries acquired over 279 s. Median vessel speed was 0.262 m s^{-1} , median capture latency was 446 ms, and GNSS quality ranged from five to six satellites with HDOP 1.55–2.35. The mission formed an approximately 24.6 m out-and-back corridor and produced one connected feature-aligned underwater transect mosaic. It also exposed the decisive limitation of the prototype: a single low-cost GNSS receiver and a six-axis IMU do not provide absolute bow heading during low-speed manoeuvres. To address the geometric part of the problem, the firmware implements manual initial heading, gyroscope propagation, guarded course-over-ground correction, interlaced strip ordering, and forward-only bounded-curvature transitions. A complete multi-strip coverage mission, however, was not field-validated. The contribution is therefore not an overstated claim of precision hydrography, but a reproducible embedded architecture, a complete field dataset, a quantitative failure analysis of heading observability, and a controlled experimental roadmap toward accurate mapping, local debris detection, and acoustic sensing.

Keywords: autonomous surface vehicle; shallow-water imaging; image mosaicking; coverage planning; embedded sensing; Lake Issyk-Kul

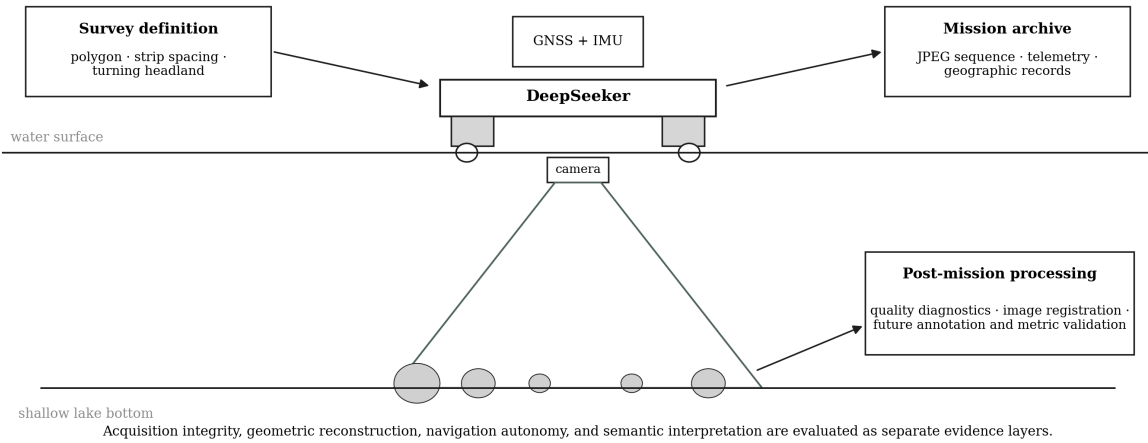


Figure 1: DeepSeeker mission concept. The platform couples survey definition, surface navigation, downward-looking image acquisition, synchronized mission logging, and post-mission analysis. The manuscript treats acquisition integrity, geometric reconstruction, navigation autonomy, and semantic interpretation as separate evidence layers.

1 Introduction

Lake margins occupy an uncomfortable space between terrestrial and marine surveying. They are often too shallow, confined, or obstacle-rich for conventional survey boats, yet they are difficult to inspect systematically by wading or diving. Manual inspection is slow, exposes people to cold water and unstable ground, and rarely reproduces exactly the same path on a later date. These limitations matter at Lake Issyk-Kul, where recreational use, coastal development, natural sediment transport, and environmental protection coexist in a narrow nearshore zone. The lake and its surrounding landscape form a UNESCO biosphere reserve, while sediment-core studies show a long-term record of anthropogenic inputs [1, 2]. A small robotic platform cannot replace chemical analysis, ecological sampling, or certified hydrographic survey. It can, however, create repeatable visual evidence, document the spatial distribution of visible objects, and provide a locally maintainable experimental base for more advanced sensors.

Autonomous and semi-autonomous surface vehicles have become practical carriers for cameras, water-quality probes, echo sounders, and acoustic receivers. Catamaran geometry is common because two separated hulls provide a broad support polygon, a central payload area, and direct differential steering. The same geometry appears in systems of very different maturity. CORAL is a 1.84 m research catamaran carrying inertial navigation, GNSS, cameras, a single-beam echo sounder, and side-scan sonar [6]. Reef Rover used a small autonomous surface vehicle and consumer cameras to acquire shallow coral-reef imagery, but its authors documented that insufficient overlap in one configuration prevented direct reconstruction until manual intervention was introduced [7]. Gogendeau et al. developed an open and reproducible multimodal ASV for acoustic tracking, bathymetry, and photogrammetry, with separate field case studies for each function and a sensor-dependent cost of several thousand dollars [8]. These systems show that a convincing research vehicle is not defined by a photograph of the platform or by the nominal presence of sensors. Its scientific value depends on synchronized acquisition, explicit coordinate frames, field validation, and error analysis.

The present project began under a different constraint. DeepSeeker was designed, constructed, wired, programmed, and tested during a short field camp using locally available components. The prototype therefore used a Main ESP32, an ESP32-S3-CAM with an OV3660 sensor, a GY-NEO6MV2 receiver, an MPU-6050, two electronic speed controllers, and two submerged thrusters. The design objective was not merely remote control. The intended mission was a lawnmower-style survey in which the catamaran would traverse parallel strips, capture overlapping underwater images, and produce a coherent visual map of the lake bottom. That objective immediately coupled three problems that are often discussed separately: vehicle guidance, underwater imaging geometry, and data integrity.

Coverage guidance is more demanding than waypoint navigation. A vessel can reach isolated coordinates while still producing an unusable image sequence. Photographic coverage requires repeatable strip spacing, along-track overlap, controlled speed, and entry into each line with a suitable orientation. The vehicle must also leave enough external space for turns, because a compact 180° pivot can destroy the very navigation information needed to complete the manoeuvre. Coverage-path planning has a long history in robotics [3, 4]. Recent work for environmental-monitoring USVs combines parallel sweep lines with Dubins connectors and validates the route in real water rather than simulation alone [5]. This is directly relevant to DeepSeeker: a wide forward-only transition may be less geometrically compact than a pivot, but it better matches the hydrodynamics of a small twin-thruster craft and preserves translational motion.

The central navigation difficulty was not a lack of code. It was a lack of absolute heading observability. The MPU-6050 measures angular rate, so yaw can be propagated only by integration and therefore drifts with bias. A single-antenna GNSS receiver reports position and course over ground, but course over ground is the direction of translation, not the orientation of the bow. It becomes unreliable or stale when speed is low, especially during a tight turn. Wind and surface current can also separate the direction of travel from the direction in which the hull points. Consequently, an algorithm may know approximately where the boat is while still not knowing which way its bow is facing. This distinction explains why increasingly complex software cannot fully compensate for an unobservable state.

Underwater imagery adds a second class of uncertainty. The number of stable visual correspondences changes with bottom texture, suspended material, sunlight caustics, camera motion, and distance to the bottom. A camera mounted behind a flat underwater window also violates the ideal pinhole model because rays refract through water, the window material, and air. Ignoring this geometry introduces systematic reconstruction error [9]. Accurate metric products require calibration in the final housing, a rigid and known camera-to-port geometry, independent scale, and check points [10, 11]. Without these elements, a connected mosaic can be a useful visual product but should not be called a metrically validated orthophoto.

Semantic detection is downstream of both navigation and imaging. Public datasets such as SeaClear and TrashCan demonstrate that submerged debris can be detected and segmented with modern neural networks [12, 13]. They also show that performance changes sharply between sites, cameras, and water conditions. A model trained on neighbouring frames from the same mission may appear accurate while failing on a new beach or on a day with different visibility. Therefore, a local YOLO pipeline requires mission-separated train, validation, and test sets. It also requires a clear distinction between a frame-level acquisition coordinate and the true coordinate of an object on the bottom.

This work is structured around evidence rather than intended capability. It asks three questions. First, can a two-controller embedded system preserve a complete sequence of high-resolution underwater images with synchronized navigation records during a real field mission? Second, what does the M0016 archive reveal about the suitability of a single GNSS receiver and a gyroscope-only yaw estimate for compact area coverage? Third, which controlled experiments are required before the imaging, acoustic, and object-detection branches support quantitative environmental claims?

The paper makes five contributions. It documents the complete embedded architecture and capture transaction; analyses all 208 frames and telemetry records from M0016; identifies low-speed heading observability as the dominant limit to lawnmower coverage; formalizes the implemented forward-only transition strategy; and converts unfinished extensions into preregistered experiments rather than retrospective claims. The result is a field-evaluated systems paper and a development baseline for a substantially stronger second-generation platform.

2 Research Context and Competitive Positioning

A competitive paper does not become strong by imitating the hardware of a mature platform. It becomes strong by matching the discipline with which mature work defines requirements, validates subsystems, and separates measured performance from design intention. The reference systems considered here illustrate four different standards of evidence.

Reef Rover is particularly relevant because its scientific value includes a documented failure. The vehicle collected imagery suitable for coral mapping, but one configuration did not maintain sufficient overlap for

automatic reconstruction [7]. The authors did not hide the failure; they used it to identify speed, depth, and camera arrangement as design variables. This is a useful precedent for DeepSeeker. The absence of a successful full lawnmower mission is not a reason to invent one. It is an opportunity to show why the original navigation stack failed and how the next experiment should isolate the cause.

CORAL demonstrates the importance of multisensor integration. Its navigation, scientific payloads, and operating modes are described as parts of one system rather than as independent modules [6]. The platform is larger and more capable than DeepSeeker, but its design principle is transferable: navigation timestamps, payload timestamps, and sensor geometry must be treated consistently. DeepSeeker approaches this requirement through a simpler dual-controller architecture in which each image receives an associated telemetry record and geotag.

The ASV of Gogendeau et al. is the most direct benchmark for research structure. The paper first defines operational requirements, then explains mechanical, electrical, and software design, and finally validates separate mission types with dedicated field surveys [8]. Its strength is not decorative design. It is the chain from requirements to hardware to field results. The current DeepSeeker dataset is narrower, but it supports the same kind of disciplined structure: requirements, implementation, complete raw archive, quantitative results, failure analysis, and next-stage validation.

Commercial BlueBoat-class platforms define a different benchmark. They provide modern GNSS, absolute heading from multiple compasses, mature autopilot software, standardized failsafes, and documented integration with sonar and communications. A first prototype built during a camp should not claim parity with such a system. Its credible competitive advantage lies elsewhere: local maintainability, inspectable code, repairable construction, and an architecture that can be upgraded incrementally. Cost remains relevant, but cost without positioning accuracy, heading observability, endurance, or validated payload performance is not a research contribution.

Table 1 therefore compares evidence rather than marketing claims. DeepSeeker already has a complete field archive and an implemented coverage planner. It does not yet have an absolute heading source, independent trajectory ground truth, calibrated photogrammetry, or validated semantic detection. The competitive research opportunity is to make those boundaries explicit and to define experiments that convert each limitation into a measurable engineering question.

Table 1: Positioning of DeepSeeker relative to representative research and commercial platforms.

Platform	Navigation and control	Primary payloads	Evidence available in the cited work
DeepSeeker	Single GNSS, six-axis IMU, manual initial yaw, differential thrust, implemented bounded-curvature coverage planner	OV3660 underwater camera; exploratory analog acoustic branch	208-image field transect with complete metadata; full grid, metric mapping, acoustic classification, and YOLO evaluation not yet validated
Reef Rover [7]	Autonomous transects on a small USV	Consumer cameras for coral mapping	Field imagery and SfM products; insufficient-overlap failure documented and discussed
CORAL [6]	GNSS/INS-supported autonomous, semi-autonomous, and assisted operation	Cameras, echo sounder, side-scan sonar	Multiple coastal field surveys and integrated sensor architecture
Gogendeau et al. [8]	Research-grade autonomous navigation with RTK-capable positioning	Acoustic tracking, bathymetry, photogrammetry	Dedicated field case study for each mission; coverage, endurance, and accuracy quantified
BlueBoat-class commercial platform	Modern GNSS, dual compasses, mature autopilot and failsafes	Modular camera, communication, and sonar integrations	Commercially documented endurance, waypoint modes, telemetry, and integration pathways

3 System Requirements and Design Rationale

The prototype was evaluated against requirements that are achievable for a first field system and relevant to later scientific use. These requirements were divided into platform, acquisition, guidance, safety, and evidence categories. The separation is important because a vehicle can satisfy one category while failing another. For example, the boat may remain stable and preserve every image while still being unable to enter the next survey strip with the required heading.

The platform had to be deployable from shore, carry the control electronics above the waterline, hold a camera below the surface, and maintain low-speed stability. The acquisition system had to save original high-resolution images locally, associate each image with navigation metadata, and continue after short storage or communication failures. Guidance had to support manual operation, straight segments, and an upgrade path toward multi-strip coverage. Safety required common-ground wiring, controlled arming, loss-of-GNSS handling, excessive-tilt detection, camera supervision, geofence logic, and a no-progress timeout. Finally, the evidence requirement demanded that design intent, firmware implementation, and field demonstration remain separate in the manuscript.

Table 2: DeepSeeker design requirements and present evidence status.

Requirement class	Requirement	Current evidence
Deployment	Shore-deployable catamaran with stable low-speed operation and protected electronics	Demonstrated in repeated field operation and M0016 photographs
Acquisition	High-resolution underwater frames stored independently of radio coverage	Demonstrated: 208 QXGA JPEGs stored locally
Synchronization	One navigation record and one geographic record for every requested image	Demonstrated: 208/208/208 correspondence
Guidance	Differential thrust, manual control, straight-line following, and extensible area planner	Manual and semi-autonomous operation demonstrated; full area coverage not demonstrated
Heading	Maintain bow orientation through low-speed turns	Not satisfied with the current GNSS + six-axis IMU stack
Mapping	Produce a connected image sequence and later a metric orthomosaic	Connected transect demonstrated; metric orthomosaic not demonstrated
Semantic sensing	Provide data structure for local debris annotation and model training	Dataset organization implemented; trained detector not yet evaluated
Acoustic sensing	Explore receiver-chain information beyond a binary range threshold	Analog access concept prepared; controlled underwater dataset not completed

Figure 2 summarizes the evidence hierarchy adopted throughout the paper. A capability is labelled field-demonstrated only when it appears in the raw mission data or source photographs. A capability can be described as implemented when it exists in supplied firmware but has not been validated in a complete mission. A planned capability is described only through an explicit protocol and expected metrics.

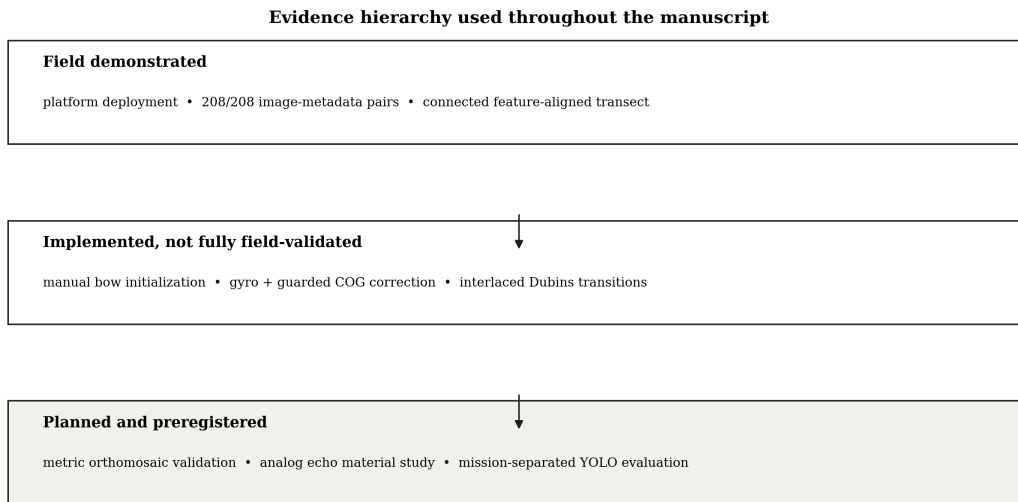


Figure 2: Evidence hierarchy used to separate field-demonstrated results, implemented but incompletely validated functions, and planned experiments.

4 Platform and Embedded Implementation

4.1 Mechanical platform and propulsion

DeepSeeker is a twin-hull surface craft with an overall length of 1130 mm and a beam of 800 mm. Two sealed PVC tubes provide buoyancy and support a reinforced wooden deck. The broad hull separation increases transverse stability and leaves a central region for the battery, control enclosure, and camera mounting. Two brushless underwater thrusters are installed at the stern and driven through independent electronic speed controllers. Equal thrust produces forward motion; a thrust difference produces yaw. A fixed bow fin was added during development to reduce lateral wandering and increase passive directional stability on straight segments.

The propulsion supply is 12 V. A regulated 5 V rail powers the controllers and sensors. The principal wiring is: GNSS TX to Main ESP32 GPIO16, GNSS RX from GPIO17, MPU-6050 SDA/SCL to GPIO21/22, left ESC signal to GPIO25, right ESC signal to GPIO2, sonar test lines to GPIO26/27, Main ESP32 TX/RX to the camera controller through GPIO14/13, and a common ground across all subsystems. The final paper does not present the prototype as optimized naval architecture. The mechanical design is a field-capable carrier whose purpose is to support sensor and autonomy experiments.

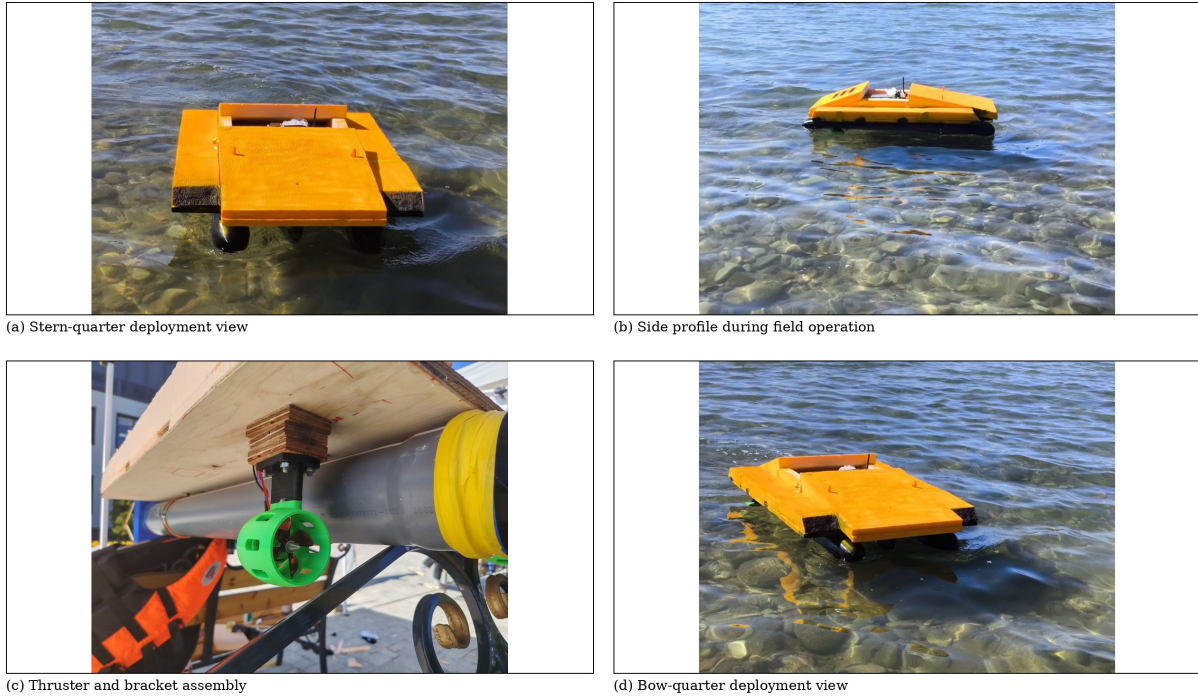


Figure 3: DeepSeeker as built and tested. The photographs are reproduced from the project archive without enhancement, background replacement, or synthetic modification.

4.2 Dual-controller architecture

The control architecture separates time-critical navigation from image encoding and storage. The Main ESP32 parses GNSS data, reads the MPU-6050, generates PWM commands for the two ESCs, executes the mission state machine, runs the path follower, and supervises failsafes. The ESP32-S3-CAM controls the OV3660 camera, writes JPEG files to microSD, stores metadata, and returns explicit capture acknowledgements.

This division addresses a practical embedded-systems problem. JPEG capture and SD-card writes can take hundreds of milliseconds and occasionally stall. If the same processor were responsible for motor output, navigation updates, web control, and image storage, a slow write could delay the control loop. The separated architecture contains this failure. The Main ESP32 can retain mission state and stop the vessel if the camera does not acknowledge a request, while the camera controller can reinitialize storage without blocking motor control.

The camera configuration recorded in firmware and confirmed by the archive is QXGA JPEG at 2048×1536 pixels, JPEG quality setting 10, 20 MHz XCLK, and local microSD storage. At initialization, several frames are discarded so that exposure and white balance settle before mission capture. Data are stored in sequential mission folders. Each requested image is linked to a row in `telemetry.csv` and to a coordinate entry in `geo.txt`.

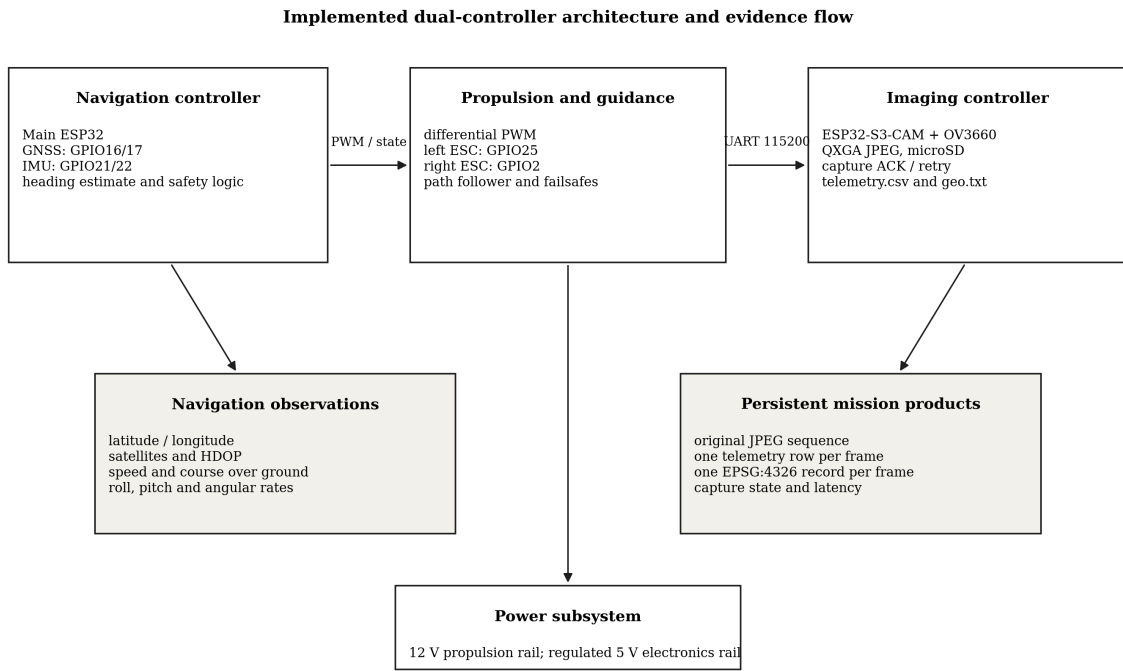


Figure 4: Implemented dual-controller architecture and evidence flow. Navigation and mission safety are separated from high-resolution image capture and persistent storage.

4.3 Capture transaction and fault containment

Communication between processors runs at 115200 baud. For each exposure, the Main ESP32 sends an image identifier and the current navigation state: UTC time, latitude, longitude, altitude, satellite count, HDOP, speed, GNSS course, roll, pitch, angular rates, total acceleration, and stability flags. The camera controller captures the image, writes the JPEG, appends a telemetry row, appends the geographic record, and returns one of three states: complete, partial, or error. Storage retry logic can reinitialize the SD card after a failure.

The transaction is deliberately auditable. A file count alone cannot establish data integrity because a frame may exist without a matching coordinate, or a metadata row may exist for a failed image. The one-to-one relationship in M0016 shows that the complete transaction was preserved for every requested frame. This is one of the strongest results of the prototype because it supports every later processing step.

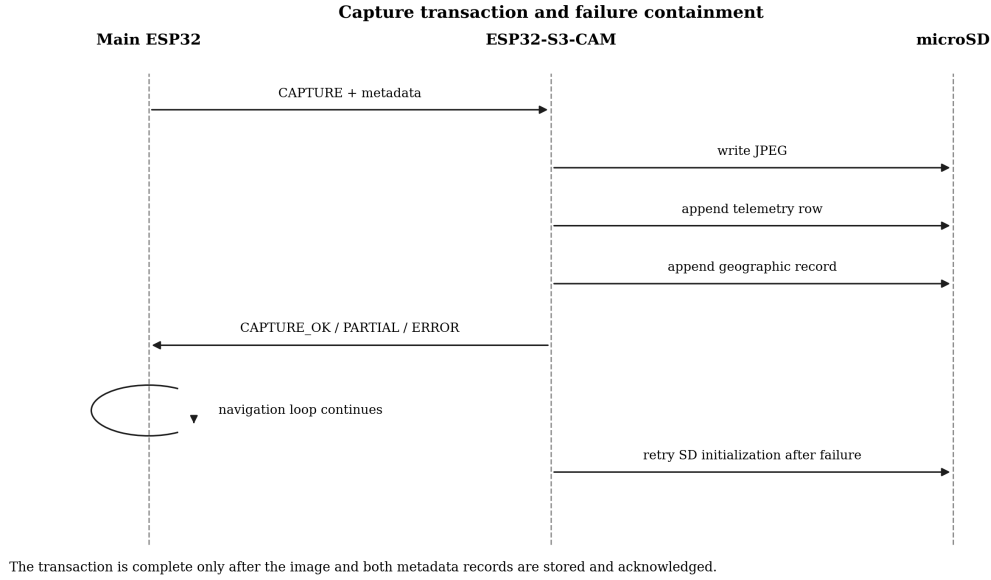


Figure 5: Capture transaction and failure containment. A request is considered complete only after the image and both metadata records are stored and acknowledged.

5 Navigation and Coverage Planning

5.1 Why the missing absolute heading matters

A surface vehicle has at least two different horizontal angles: bow heading ψ , describing the orientation of the hull, and course over ground χ , describing the direction of translational motion. These angles are similar only when sideslip and external disturbances are small. During wind, current, or asymmetric thrust, they can differ. More importantly, course over ground is undefined or unstable when the vessel is almost stationary.

The MPU-6050 provides a yaw rate ω_z , not absolute yaw. After estimating a gyro bias $\hat{b}_g$, the heading can be propagated as

$$\hat{\psi}_k = \text{wrap}_{180} [\hat{\psi}_{k-1} + (\omega_{z,k} - \hat{b}_g)\Delta t]. \quad (1)$$

Any residual bias accumulates. The firmware therefore uses a guarded complementary correction toward GNSS course over ground when speed, freshness, and plausibility conditions are satisfied:

$$\hat{\psi}_k \leftarrow \hat{\psi}_k + \alpha \text{wrap}_{180}(\chi_{\text{GNSS}} - \hat{\psi}_k), \quad 0 < \alpha \ll 1. \quad (2)$$

This method can reduce drift on straight moving segments, but it cannot create an absolute orientation measurement during a low-speed pivot. It is a heading surrogate, not a full attitude solution.

Before a mission, the operator physically points the bow toward the first target and invokes a bow-lock routine. During the stationary interval, the code estimates gyroscope bias and accepts the geometric bearing to the first target as the initial heading. This procedure is practical for a prototype, but it means that the autonomy is operator-initialized. A rigorous description is therefore semi-autonomous or autonomy-capable, not fully autonomous.

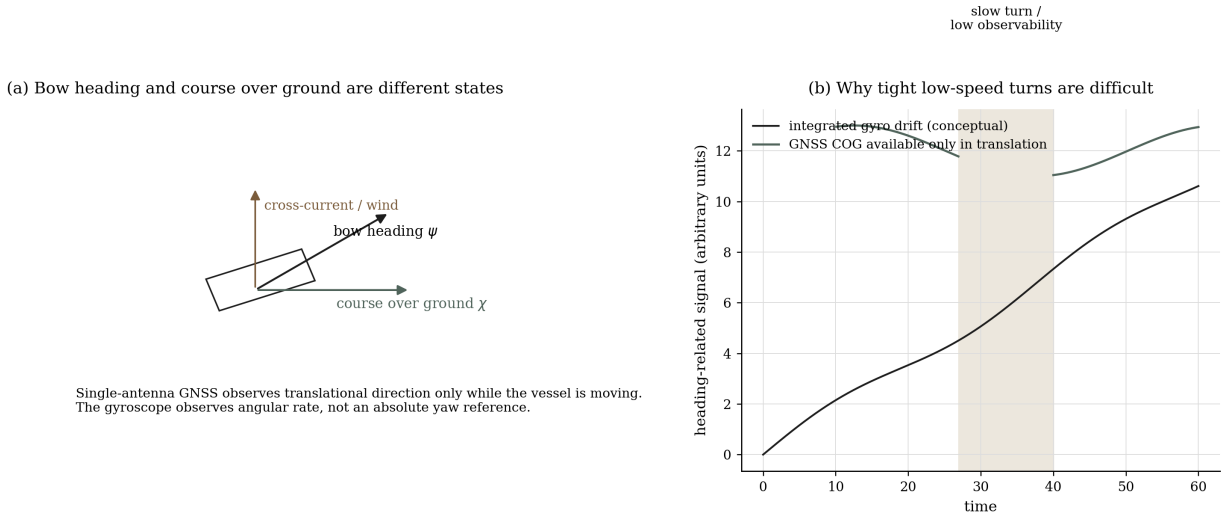


Figure 6: Heading observability. (a) Bow heading and course over ground represent different states under wind or cross-current. (b) A tight low-speed turn removes the translational information needed for GNSS course correction while gyro integration continues to drift.

5.2 Interlaced coverage and the teardrop transition

The intended survey mode is a set of parallel photographic strips inside a user-defined quadrilateral. Default strip spacing is 1.5 m, nominal capture interval is 1.1 s, and the default minimum turn radius is 2.0 m. The values are software defaults rather than validated physical optima. Strip spacing should eventually be derived from calibrated camera footprint and required side overlap; capture timing should be derived from speed and required forward overlap.

An adjacent-strip lawnmower pattern demands a nearly immediate reversal at the end of each line. For a small differential-thrust vessel, this can require one propeller to reverse or the boat to rotate almost in place. Such a manoeuvre slows the vessel, magnifies disturbance effects, and removes a reliable GNSS course. The project therefore implemented a wider external turn. The current strip is extended beyond the photographic region, the vessel follows a forward-only loop in an external headland, and it enters the next selected strip already aligned.

The implementation is more rigorous than a manually drawn arc. For the end pose of the current strip and the entry pose of the next strip, the code evaluates the six canonical Dubins path families: LSL, RSR, LSR, RSL, RLR, and LRL. Each feasible candidate is discretized. Paths that re-enter the photographic rectangle are rejected, and the shortest remaining candidate is selected. The lane order is interlaced: even-indexed strips are visited first, followed by odd-indexed strips. This increases the lateral distance available between consecutively visited lines.

The approach is physically plausible and consistent with current coverage-planning literature [5]. It is not yet a validated performance claim. The default radius must be measured against actual speed, battery voltage, loading, wind, and thrust asymmetry. A future comparison must quantify entry-heading error, cross-track error, transition time, minimum speed, energy use, and completion rate for pivot turns and several Dubins radii.

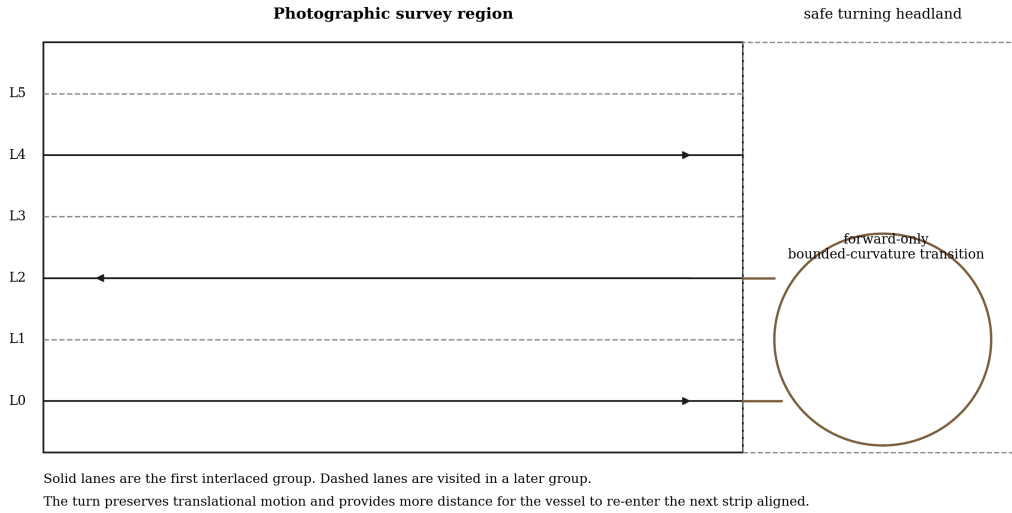


Figure 7: Implemented interlaced coverage concept. Solid lanes represent the first lane group; dashed lanes are visited later. The forward-only bounded-curvature turn is performed in a safe external headland rather than inside the photographic region.

6 Field Campaign and Analysis Methods

6.1 Mission M0016

Mission M0016 was conducted in shallow water at Lake Issyk-Kul on 17 July 2026. The raw archive contains 208 consecutively numbered JPEG files, 208 rows in `telemetry.csv`, and 208 entries in `geo.txt`. No image indices are missing. The first recorded exposure occurred at 03:41:25 UTC and the last at 03:46:04 UTC, giving an acquisition span of 279 s.

M0016 was an operator-assisted out-and-back image transect rather than a complete autonomous grid. That distinction is central to the interpretation. The mission is suitable for evaluating acquisition integrity, capture latency, GNSS behaviour, attitude, raw-image variability, and connected image registration. It is not suitable for measuring multi-strip coverage success or the performance of the implemented teardrop transition.

6.2 Trajectory processing

Latitude and longitude were converted to local east-north displacement using a small-area tangent approximation referenced to the first fix. Principal-component analysis was used descriptively to identify the dominant transect axis and estimate its span. PCA was not used as ground truth. The mission was divided into outbound frames 1–121, reversal frames 122–129, and return frames 130–208 based on the reversal in progression along the principal axis.

Summing point-to-point GNSS distances would overestimate travelled distance because metre-scale jitter is comparable to the separation between consecutive fixes. The analysis therefore reports bounding dimensions, principal-axis span, and the qualitative structure of the track instead of an apparent path length. Consecutive repeated coordinates were counted because they reveal frames captured before a new GNSS fix was available.

6.3 Image diagnostics

All image metrics were calculated from the original QXGA files without enhancement or recolouring. Grayscale standard deviation was used as a descriptive measure of contrast. Variance of the Laplacian was used as a descriptive edge-sharpness measure. Shannon entropy was used to describe texture information. These quantities are not universal underwater image-quality scores and do not directly measure photogrammetric accuracy.

The supplied mosaic was evaluated as a connected visual product. Because the reconstruction code, detector

and matcher settings, underwater calibration, control points, and independent scale were not part of the archive, the paper does not report ground-sampling distance, reprojection error, map accuracy, or registered-frame percentage. The product is described as a feature-aligned transect mosaic rather than an orthophoto.

7 Results

7.1 Complete acquisition and timing performance

The M0016 archive contains 208 images, and every image has a matching telemetry row and geographic record. This 208/208/208 correspondence is the primary systems result. It confirms that the two-controller transaction survived the entire field mission without losing the logical relationship between image and navigation data.

Median capture latency was 445.5 ms. The 95th percentile was 516.6 ms, and the maximum was 1124 ms. JPEG size ranged from 81.8 to 120.1 KiB, with a median of 100.5 KiB. The UTC field has one-second resolution, so timing cannot be interpreted as a high-precision frame clock. Among 207 consecutive intervals, the median recorded gap was 1 s and the 95th percentile was 2.7 s. Thirty-four image pairs were recorded within the same UTC second, and five pauses lasted at least four seconds; the longest was 33 s. Sequence numbering remained continuous across the pauses.

These observations support two conclusions. First, local storage and explicit acknowledgements provided robust transactional logging. Second, the capture cadence was not uniform enough to be treated as a fixed frame rate. A future survey controller should trigger images by travelled distance and camera footprint rather than by a fixed timer alone.

7.2 Trajectory and GNSS behaviour

Median speed at image capture was 0.262 m s^{-1} , the mean was 0.283 m s^{-1} , and the 95th percentile was 0.628 m s^{-1} . Satellite count remained between five and six. HDOP ranged from 1.55 to 2.35, with a median of 1.56. The local GNSS positions occupied a bounding box approximately 9.2 m east-west by 24.4 m north-south. The dominant principal axis spanned approximately 24.6 m, confirming that the mission was a long corridor rather than area coverage.

Figure 8 shows the capture positions reconstructed independently from the raw telemetry. The outbound and return passes are visibly separated, with a compact reversal at the far end. The median cross-axis separation between the two pass groups was approximately 4.1 m, but this value is derived from the same consumer GNSS and is not a surveyed lane-spacing measurement.

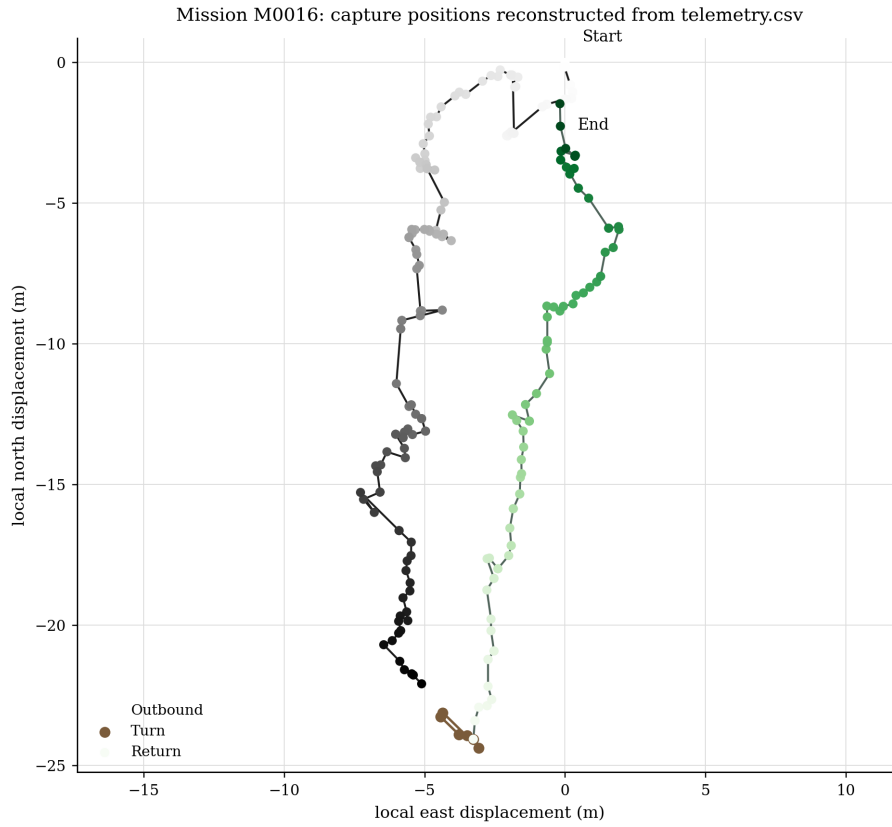


Figure 8: Mission M0016 capture positions reconstructed from `telemetry.csv`. Points are GNSS acquisition positions, not a surveyed ground-truth trajectory.

Two telemetry artefacts explain the navigation limitation more directly than the apparent track shape. First, 34 consecutive image pairs share identical latitude and longitude, showing that image capture sometimes occurred faster than GNSS position updates. Second, `course_deg` contains only one unique value, 115.24° , across all 208 rows. The physical track clearly reverses, so the field cannot represent a valid measured bow heading or an updated course for the mission. This stale or unrefreshed value removes the correction signal on which the magnetometer-free heading estimate depends.

7.3 Attitude and capture-state behaviour

All frames satisfied the firmware’s broad $\pm 18^\circ$ level criterion. Only 38.0% satisfied the stricter short-window stability flag. Median absolute roll was 0.51° , with a 95th percentile of 1.63° . Median absolute pitch was 1.93° , with a 95th percentile of 3.90° . The platform therefore remained within a moderate attitude envelope, but high-frequency motion often exceeded the stricter gyro and acceleration condition.

The firmware allowed deadline captures when waiting for a more stable state would risk losing overlap. This choice favoured sequence completeness over instantaneous sharpness. It was effective for acquisition integrity, but it should be studied explicitly in future work by relating the stability flag to keypoint count, inlier ratio, and registration failure.

7.4 Synchronized telemetry and visual conditions

Figure 9 shows speed, capture latency, roll and pitch, and standardized image diagnostics for the complete mission. The highlighted interval corresponds to the principal reversal. The speed trace confirms that the vehicle operated mainly in a low-speed regime. Capture latency remained near the median for most frames but included occasional spikes. Roll and pitch stayed moderate. Edge sharpness and gray-level contrast declined during the middle part of the sequence and recovered later.

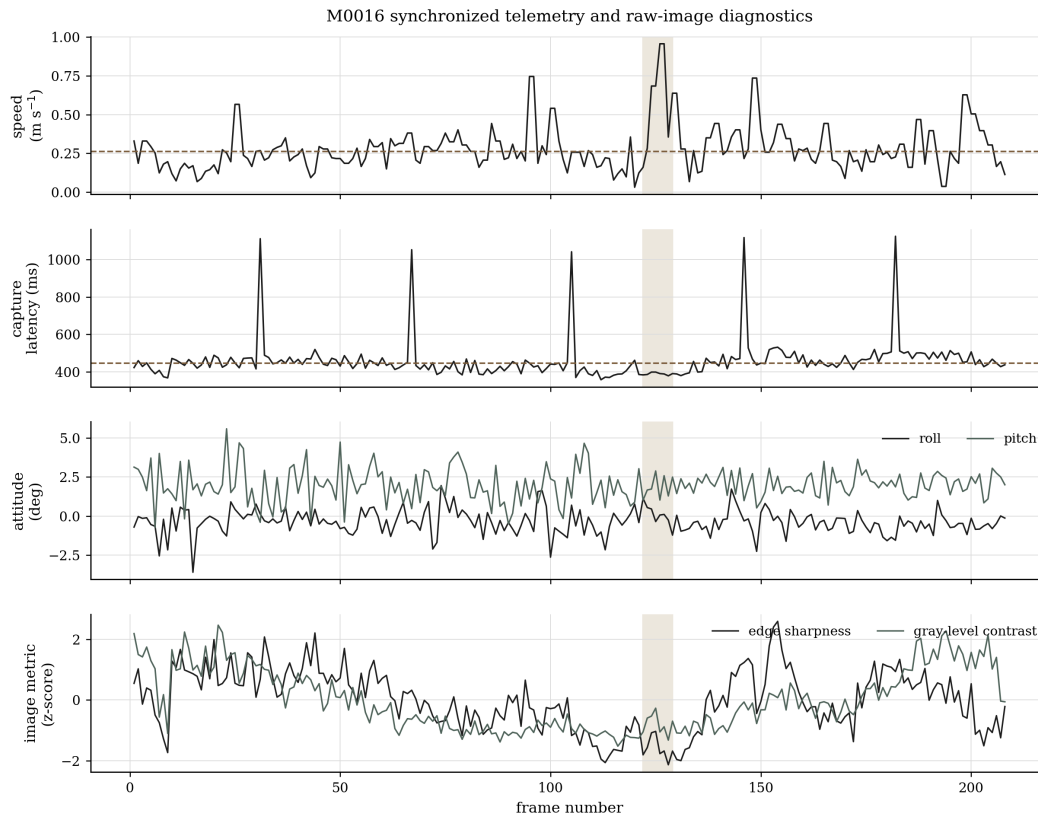


Figure 9: Synchronized telemetry and raw-image diagnostics for all 208 frames. The shaded interval contains the principal track reversal. Image metrics are standardized descriptive quantities, not calibrated mapping accuracy.

Full-resolution Laplacian variance had a median of 2.879 and ranged from 1.460 to 4.570 in the adopted OpenCV scale. Grayscale standard deviation had a median of 25.22 intensity levels and ranged from 11.13 to 55.12. Shannon entropy had a median of 6.530 bits and ranged from 5.376 to 7.361 bits. These values agree with direct visual inspection. Early frames show high-contrast cobbles and sunlight caustics. The middle portion contains smoother, lower-contrast sediment and weaker texture. Later frames recover coarser rock texture.

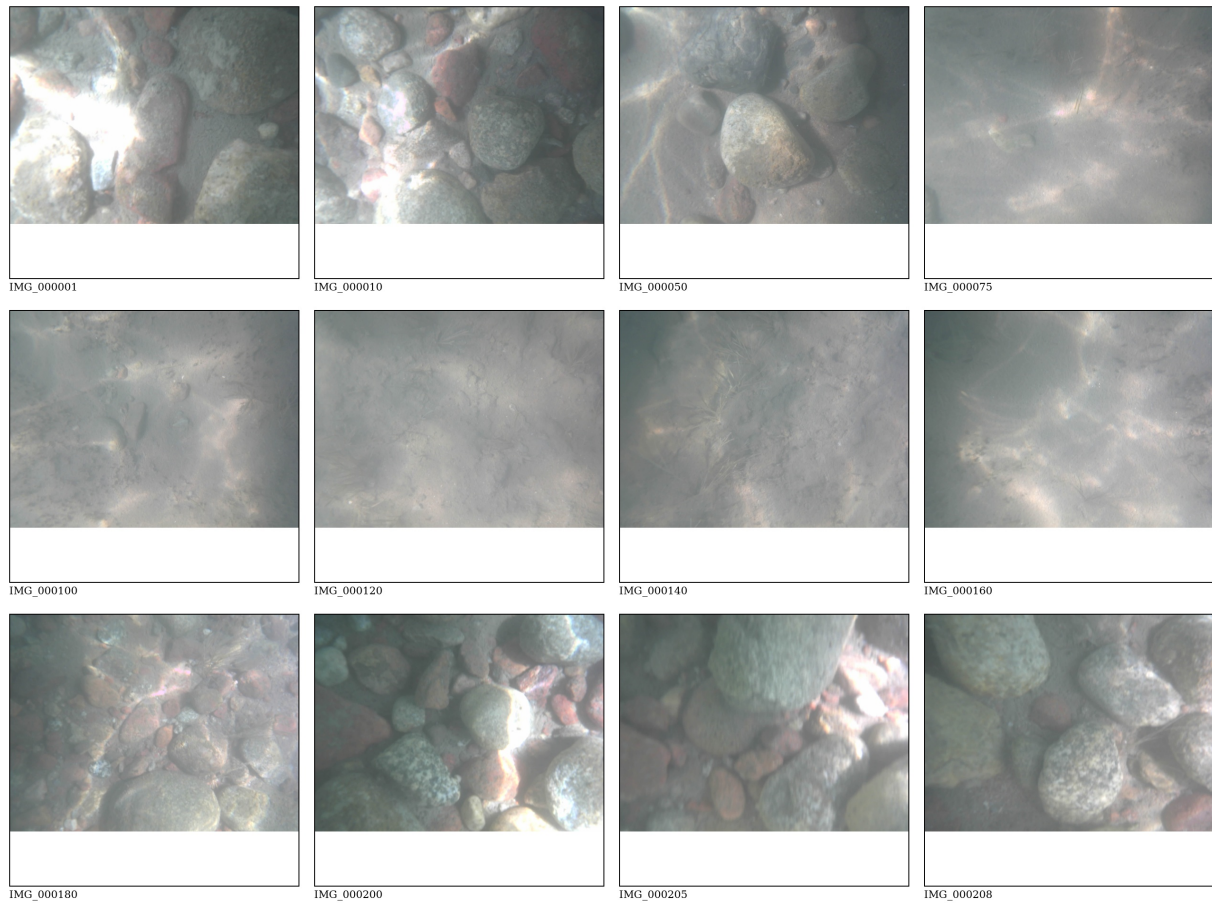


Figure 10: Representative original frames spanning the complete M0016 sequence. The panel shows the transition from coarse cobble texture to lower-contrast sediment and back to stronger rock texture.

7.5 Connected image registration

The supplied image product forms one connected visual transect. This is a meaningful field result: the sequence retained sufficient visual continuity to bridge a period of reduced texture and assemble a single corridor. The irregular external boundary and visible local distortion show that the product is not a finished map. No check points, independent scale, refractive camera calibration, or reconstruction log were supplied. Consequently, the manuscript does not claim metric orthorectification.

The original team diagnostic, connected mosaic, and raw/enhanced comparison are retained in source form in Figure 11. Enhancement improves local contrast and colour separation to the eye, but its contribution to registration is unknown. A controlled comparison must report keypoints, accepted matches, RANSAC inlier ratio, reprojection error, and connected-component count for raw and enhanced processing branches.

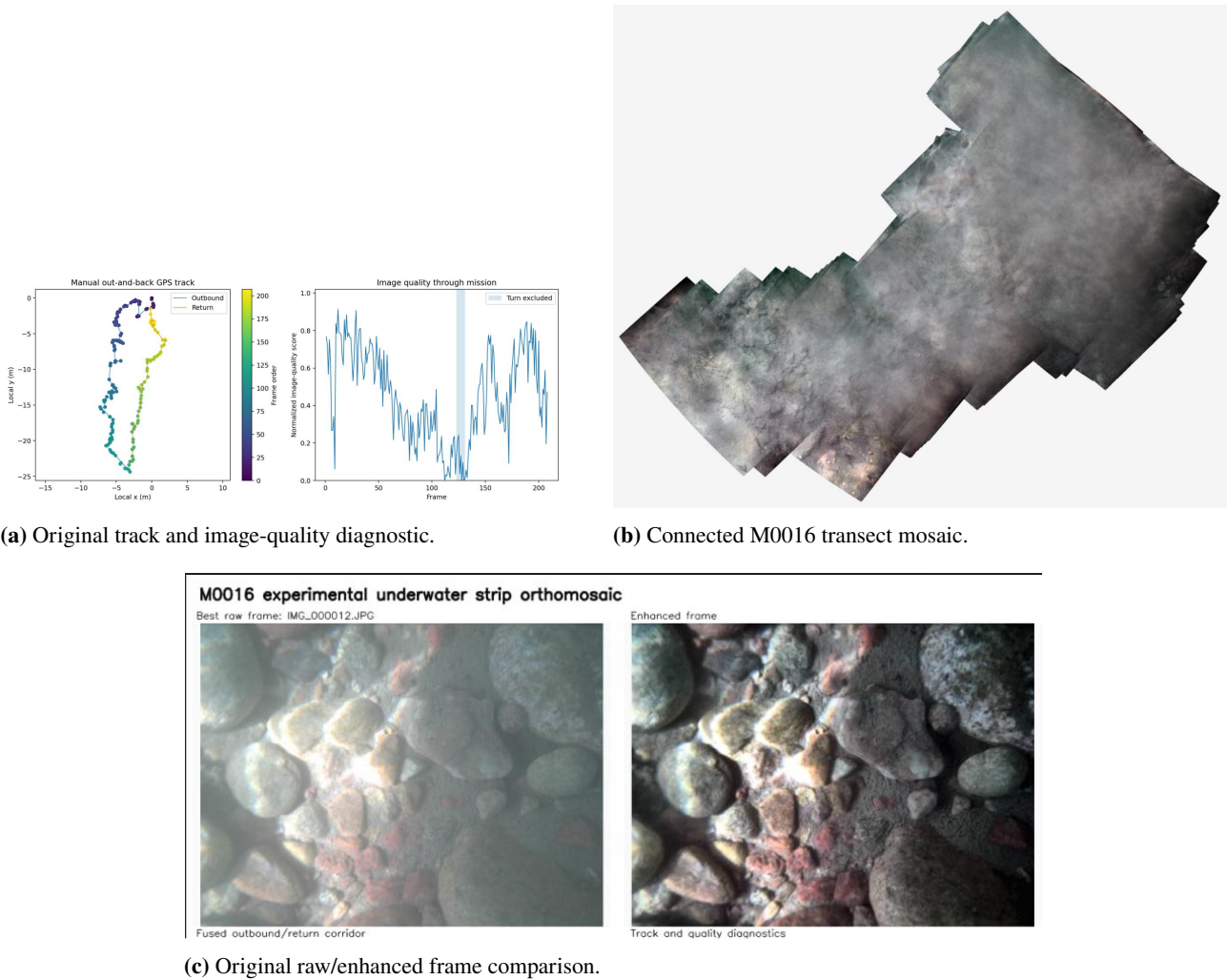


Figure 11: Original mission outputs reproduced unchanged. They support a connected feature-aligned transect, not a metrically validated orthophoto.

Table 3: Quantitative summary of mission M0016.

Metric	Observed value
Complete image–telemetry–geotag triplets	208 / 208 / 208
Acquisition span	279 s
Image dimensions	2048 × 1536 pixels
Median / mean / 95th-percentile speed	0.262 / 0.283 / 0.628 m s ⁻¹
GNSS satellites	5–6, median 6
HDOP	1.55–2.35, median 1.56
Median / 95th-percentile / maximum capture latency	446 / 517 / 1124 ms
Median JPEG size	100.5 KiB
Frames satisfying strict stability flag	38.0%
Frames satisfying broad level criterion	100.0%
Principal-axis span	24.6 m
Repeated consecutive GNSS fixes	34 pairs
Unique values in logged course field	1
Connected supplied image product	1 feature-aligned transect

8 Discussion

8.1 The strongest contribution is acquisition integrity

The platform's clearest success is the complete synchronized archive. Every requested image has a matching navigation row and coordinate record. For an ESP32-class field system, this is not a trivial administrative detail. It demonstrates that high-resolution capture, SD-card storage, motor control, and mission logging were separated effectively enough to survive the full run. The architecture gives future experiments a reliable data foundation. Better GNSS, a new camera, an altimeter, or a semantic detector can be evaluated only if the underlying mission record remains complete.

This result also changes how affordability should be discussed. The scientific advantage is not simply that inexpensive components were used. It is that the system is inspectable, repairable, and organized around explicit transactions. A locally maintainable architecture can remain less expensive than a commercial survey platform while still adopting a better GNSS receiver, dual-antenna heading, calibrated optics, and a true underwater altimeter.

8.2 The central failure is state observability, not software complexity

The repeated development difficulty is often summarized as “there was no compass.” A more precise statement is that the system lacked an absolute yaw observation during the manoeuvres in which heading mattered most. This is a deeper result. The gyroscope supplied angular rate and the GNSS supplied approximate position, but neither supplied a stable bow direction at low speed. The stale course field in M0016 provides direct evidence that the correction pathway was unavailable in the field data.

This finding explains why a complete lawnmower route could not be recovered merely by writing more code. A path planner can generate a mathematically valid route, and a controller can calculate a desired heading, while the state estimator remains unable to determine the actual heading. Adding control gains, waypoints, or more complex logic cannot remove this ambiguity. The next version should therefore prioritize sensing before controller complexity.

A dual-antenna GNSS heading solution is particularly attractive because it can provide absolute orientation without translational speed and is less vulnerable than a magnetometer to currents and magnetic materials near the propulsion system. A carefully mounted and calibrated nine-axis IMU is a lower-cost alternative, but its placement, hard-iron calibration, soft-iron calibration, and dynamic validation must be reported. RTK positioning is a separate upgrade: it improves position but does not by itself solve bow heading unless the receiver uses an antenna baseline.

8.3 Why the external turn remains a strong research direction

The implemented teardrop transition is a sound response to the interaction between vehicle dynamics and navigation sensing. A pivot turn is spatially compact but creates the worst conditions for course-over-ground correction. A broad forward-only turn takes more space but keeps the craft moving, avoids a motor reversal, and allows the hull to settle before it reaches the next image line. The planner's enumeration of all six Dubins families and rejection of paths that cross the photographic region are technically meaningful features.

The current paper deliberately stops before claiming that the turn improves performance. That claim requires a repeated field comparison. The minimum radius should not be selected by intuition alone. It should be measured as a function of forward command, differential command, battery voltage, payload, and wind. A strong experiment would compare a pivot, a conventional close U-turn, and several bounded-curvature radii. Outcomes should include actual radius, entry-heading error, cross-track error over the first metres of the next strip, minimum speed, transition duration, energy, and success rate. The safe headland must also be defined as a physical obstacle-free polygon, not merely as empty space outside the image region.

8.4 The visual result is credible because the claim is narrow

M0016 supports connected underwater mosaicking. It does not support a metric orthophoto. This distinction is essential. A visually coherent image can contain scale drift, local warping, and systematic refractive error. The current output is useful for qualitative inspection and for demonstrating overlap, but it cannot yet provide defensible object coordinates or area measurements.

A metric experiment needs underwater calibration in the final housing, a rigid camera mount, known scale bars or surveyed bottom targets, a measured camera-to-bottom distance, and a fully documented reconstruction pipeline. The pipeline should report detector, descriptor, matching rule, geometric model, RANSAC threshold, accepted-frame fraction, inlier ratio, and reprojection error. Check-point RMSE should be reported separately from fit error. Only after those measurements should the term orthomosaic be used.

The timing data also suggest an algorithmic improvement. At the median speed and 1.1 s nominal interval, the vessel travels about 0.29 m between requested exposures. That number is meaningful only if camera footprint and bottom distance are known. A better trigger computes the desired along-track step from footprint L_f and forward overlap O_f :

$$\Delta x = L_f(1 - O_f), \quad T = \frac{\Delta x}{v}. \quad (3)$$

The line spacing should be calculated similarly from cross-track footprint and side overlap. This makes the acquisition geometry adapt to speed and altitude rather than relying on fixed time and distance defaults.

8.5 YOLO and acoustic classification are future experiments, not decorative features

The project is structurally prepared for local object detection because every frame belongs to a named mission and carries acquisition metadata. That is useful, but it is not model performance. A defensible dataset should use retrievable targets, define classes before annotation, and split data by mission or site. Adjacent frames must not be divided randomly across train and test sets because they contain nearly identical scenes. Public underwater datasets can initialize the model, but the final evaluation must use unseen Issyk-Kul missions and report precision, recall, mAP50, mAP50–95, inference speed, and failure modes by visibility condition.

The analog ultrasonic branch is similarly promising only under a narrow hypothesis. Accessing the receive chain before the module's binary threshold may reveal waveform or envelope features. Echo amplitude can depend on acoustic impedance, but also on distance, angle, target size, roughness, coupling, bubbles, temperature, and gain. A single analog voltage cannot identify material. The correct experiment holds geometry constant, tests repeated targets at several ranges and angles, extracts time of flight, peak amplitude, energy, width, and spectral features, and evaluates classifiers with complete range-angle groups held out. A negative result remains scientifically useful if within-class variability exceeds between-material separation.

8.6 Broader significance for locally developed robotics

The domestic contribution of DeepSeeker is not a claim that imported components became a wholly indigenous sensor. The contribution is that mechanical integration, wiring, communication protocol, mission logic, field workflow, and failure diagnosis were carried out by the team in Kyrgyzstan. This creates local engineering capability and a platform that can be repaired and extended without replacing the entire system.

For other student and early-stage robotics teams, the paper offers a general lesson. A sophisticated route planner does not compensate for an unobservable state. A connected mosaic does not prove orthorectification. A GPS coordinate attached to a frame does not equal the coordinate of an object on the bottom. A prepared annotation workflow does not prove neural-network accuracy. Keeping these layers separate produces a stronger paper because every claim can be traced to a specific experiment.

9 Preregistered Validation Programme

The next campaign should be designed as a sequence of controlled studies rather than as another general demonstration. Each study should isolate one uncertainty and preserve the original data required for independent

reanalysis.

9.1 Navigation and turn validation

A rectangular test area should be established with independently surveyed reference points. At least five straight-line trials and five turns should be completed for each heading configuration: gyro propagation after bow lock, gyro plus guarded GNSS course correction, calibrated magnetometer, and dual-antenna heading if available. Pivot turns should be compared with bounded-curvature radii of 1.5, 2.0, 2.5, and 3.0 m at several forward commands. Primary outcomes are cross-track RMSE, 95th-percentile cross-track error, entry-heading error, minimum speed, transition time, path length, battery energy, and completion rate.

9.2 Metric optical survey

The camera should be calibrated underwater in the final housing. Focus and camera-to-port position must remain fixed. Scale bars and independent check points should be placed on the bottom. Longitudinal trigger distance and strip spacing should be computed from the calibrated footprint and target overlaps. At least three complete grids should be repeated under comparable conditions. The analysis should report registered-image fraction, connected components, tie-point inlier ratio, median reprojection error, check-point RMSE, ground-sampling distance, and uncovered-area fraction.

9.3 Analog echo-signature study

The analog tap should be buffered and protected before ADC sampling. Equal projected-area targets of aluminium or steel, PVC, waterlogged wood, and stone should be tested at 0.30, 0.50, 0.75, and 1.00 m and at incidence angles of 0° , 15° , and 30° . At least 20 repetitions per cell are required. Water temperature, target dimensions, turbidity, gain, and mounting geometry must be recorded. Features should include time of flight, peak amplitude, integrated energy, pulse width, rise time, decay time, and frequency-band energy where sampling permits. Evaluation should hold out complete range-angle groups.

9.4 Local debris detector

Only retrievable, non-polluting targets should be deployed. Initial classes may include plastic container, metal-like debris, wood, rope or textile, and natural stone/background. Data should span several missions, bottom types, depths, and visibility conditions. Train, validation, and test partitions should be separated by mission or site. Results should report class counts, annotation policy, precision, recall, mAP50, mAP50–95, confusion matrix, inference speed, and performance stratified by depth and visibility.

Table 4: Minimum next-stage validation matrix.

Experiment	Controlled variables	Primary outcomes
Straight-line guidance	Heading source, speed command, wind, payload	Cross-track RMSE, 95th-percentile error, completion rate
Turn comparison	Pivot vs. 1.5–3.0 m radii, command level, battery state	Actual radius, entry-heading error, minimum speed, energy, success
Complete grid	Strip spacing, overlap, heading source, camera height	Coverage gaps, registered-frame fraction, mission duration, map geometry
Underwater calibration	Housing geometry, focus, calibration target	Intrinsics, refractive parameters, reprojection residuals
Echo signatures	Material, distance, angle, temperature, gain	Macro-F1, balanced accuracy, confusion matrix, uncertainty
Debris detection	Class, mission, visibility, depth, bottom type	Precision, recall, mAP, inference speed, domain-shift failure

10 Limitations

The current evidence is based on one deeply analysed mission. M0016 provides a complete archive, but one mission cannot establish general reliability across wind, wave, depth, lighting, and turbidity conditions. The GNSS track is not independent ground truth, so cross-track accuracy and pass separation cannot be validated metrically. The supplied mosaic lacks reconstruction logs, underwater calibration, and check points. The implemented coverage planner was inspected in firmware but not evaluated in a complete grid. The acoustic and neural-network branches remain protocols rather than results.

The manuscript also avoids several tempting but unsupported claims. RGB imagery cannot determine heavy-metal concentration. A surface GNSS coordinate is not the exact coordinate of an object on the bottom. A waterproof ultrasonic range module is not automatically a bathymetric sounder. A connected image mosaic is not automatically an orthophoto. These limitations define the present scope and protect the credibility of the measured results.

11 Conclusion

DeepSeeker demonstrates that a locally developed, resource-constrained catamaran can collect a complete synchronized underwater image archive during a real shallow-water mission. M0016 produced 208 QXGA images, 208 telemetry rows, and 208 geographic records over 279 s. Median capture latency was 446 ms, median speed was 0.262 m s^{-1} , and the sequence produced one connected feature-aligned transect. These results establish acquisition integrity and field readiness of the embedded architecture.

The same archive identifies the decisive obstacle to full area coverage. The logged course did not update through the physical reversal, and the six-axis IMU provided no absolute yaw reference. The system could therefore propagate heading on straight moving segments but could not guarantee bow orientation during a compact low-speed turn. This is a state-observability problem rather than a shortage of software logic.

The supplied firmware contains a credible geometric response: manual initial heading, gyro propagation, guarded GNSS correction, interlaced lane ordering, and forward-only bounded-curvature transitions. The next step is not to describe this implementation more confidently. It is to validate it against pivot turns, independent position and orientation reference, repeated missions, and explicit metrics.

The project is competitive when positioned accurately. It is not yet a replacement for mature hydrographic or photogrammetric platforms. It is a complete, inspectable, locally maintainable experimental system that has already produced a high-quality raw field dataset and a clear engineering diagnosis. By preserving the evidence boundary and preregistering the next experiments, DeepSeeker provides a rigorous foundation for precise coverage guidance, calibrated shallow-water mapping, mission-separated debris detection, and controlled acoustic research at Lake Issyk-Kul.

Data and Code Availability

The analysed field archive is mission M0016 and contains 208 original JPEG images, `telemetry.csv`, and `geo.txt`. The firmware examined for this manuscript includes the Main ESP32 autonomy code with interlaced bounded-curvature planning and the ESP32-S3-CAM mission-logging code. A public release should preserve original files, checksums, reconstruction scripts, calibration files, and a machine-readable description of later annotations.

Acknowledgements

The authors thank the Young Scientists Camp organizers, project mentors, and participants who supported construction and field testing at Lake Issyk-Kul.

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